

Data Acquisition, Analysis, and Network Visualization of E-Commerce Platforms

Women Fashion Products Project

Course	DSAI 103
Project	E-Commerce Data Acquisition and Network Visualization
Domain	Women Fashion Products from Amazon, eBay API, and a scraping sample site
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Deliverable	PDF report submitted with the complete source code

A clear report explaining the implemented stages: collecting women fashion products, cleaning and preparing the dataset, building a NetworkX graph, visualizing trends using charts and heatmaps, adding point cloud analysis, and creating a final GUI for exploration.

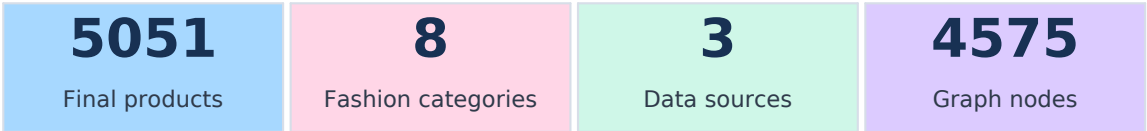
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1. Project Overview

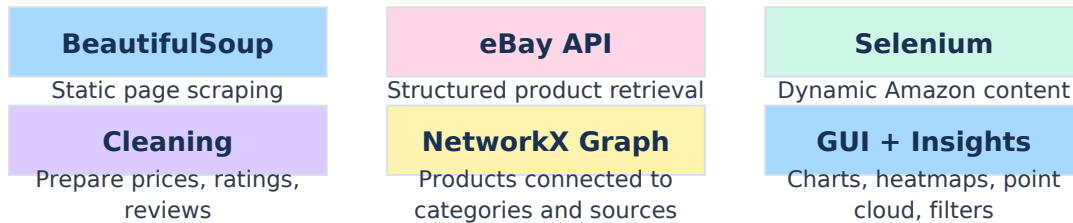
This project focuses on online shopping platforms in the women fashion domain. The aim is to collect product data from different acquisition techniques, process the data into a clean structure, and then analyze relationships and trends using graphs and visualizations.

The collected products cover categories such as dresses, tops, bottoms, outerwear, bags, shoes, modest wear, and swimwear. The project is not a data mining or search engine project; it is mainly about data acquisition, processing, network construction, visual analysis, and a final GUI for exploring the collected e-commerce data.



2. System Workflow

The notebook follows the course requirements step by step. The workflow starts with collecting data from multiple sources, then cleaning the collected records, then creating a product relationship graph, and finally producing visualizations, point cloud analysis, and an interactive GUI.



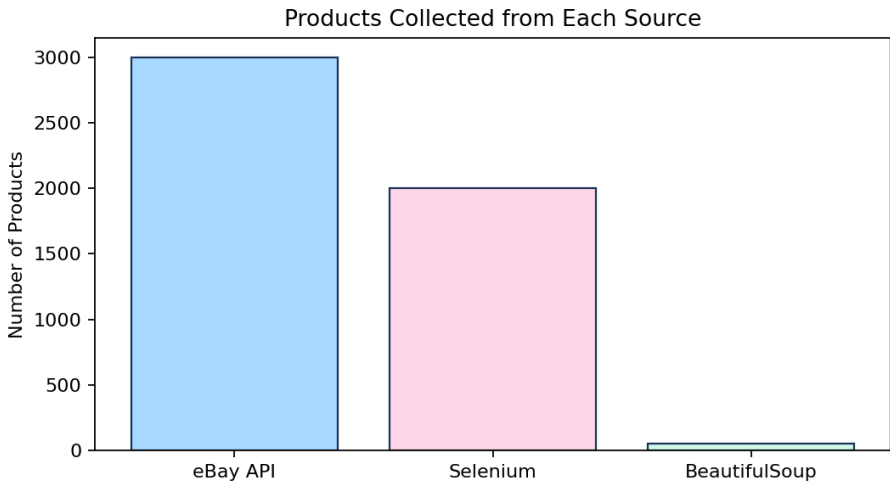
Requirement coverage:

Course requirement	Implemented in the notebook
Web scraping	BeautifulSoup was used to scrape product names, prices, images, links, and categories.
APIs	eBay API was used to collect structured women fashion product data.
Dynamic content	Selenium was used to collect Amazon Egypt products, ratings, and review counts.
Preprocessing	The dataset was cleaned, converted into numeric format, and duplicates were checked.
Graph construction	NetworkX graph was created using product-category-source relationships.
Visualization	Charts, NetworkX graph, heatmap, and point cloud analysis were included.
Final GUI	ipywidgets and Plotly were used to make simple interactive exploration tools.

3. Data Acquisition

Three different data acquisition methods were used, matching the project brief. Each method collected real product information and stored it in a dataframe before combining all records into one final dataset.

Method	Source	Collected	Main fields
BeautifulSoup	Scraping course e-commerce pages	51 products	Name, Price, Link, Image, Category, Source
API	eBay API	3000 products	Name, Price, Currency, Link, Image, Category, Source
Selenium	Amazon Egypt search pages	2000 products	Name, Price, Link, Image, Category, Rating, Reviews



The Selenium part was important because Amazon pages contain dynamic elements, and this helped collect extra values such as product ratings and review counts when they were available.

4. Data Cleaning and Preprocessing

After collecting the data, all dataframes were combined into one dataframe. The notebook then kept the needed columns only and cleaned the text and numeric values so that the dataset could be used in analysis and visualization.

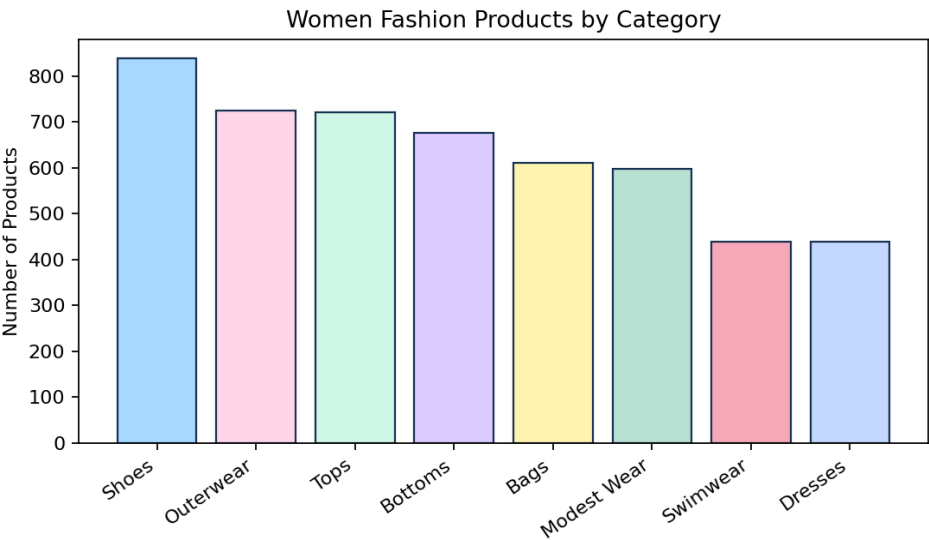
Text Fields Name, Link, Image, Category, and Source were stripped from extra spaces.	Price EGP, USD, dollar signs, commas, and extra spaces were removed before numeric conversion.	Rating Text such as 'out of 5 stars' was converted into numeric rating values.
Reviews Parentheses and commas were removed, then values were converted to integers.	Duplicates Duplicate products were checked using Name and Link.	Missing Values Missing ratings were kept because not all sources provide rating information.

Cleaning output from the notebook:

Check	Result
Total before cleaning	5051 products
Total after cleaning	5051 products
Duplicate rows	0
Shape after removing duplicates	(5051, 8)
Missing prices	10
Missing ratings	3473, because ratings were mainly available from Selenium only

5. Dataset Summary

The final dataset is focused only on women fashion products. It includes eight categories and three data sources. This gives the project enough variety for category analysis, source comparison, network analysis, heatmaps, and point cloud visualization.



Category	Products
Shoes	839
Outerwear	726
Tops	721
Bottoms	676
Bags	611
Modest Wear	599
Swimwear	440
Dresses	439

6. NetworkX Graph Construction

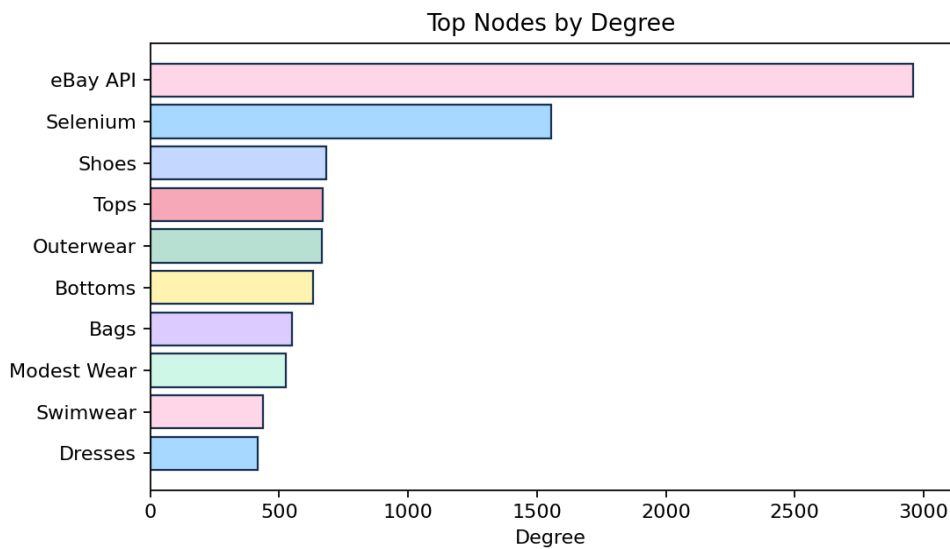
A graph was constructed using NetworkX to represent relationships in the collected e-commerce data. Each product was added as a product node. Each category and each source were also added as nodes. Then every product was connected to its category and to its source.

Product Node Example: product name Product -> Category edge Shows product classification	Category Node Example: Shoes Product -> Source edge Shows acquisition source	Source Node Example: eBay API Graph attributes Price, rating, reviews
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Graph metric	Value from notebook
Number of nodes	4575
Number of edges	9141
Node types	Product, Category, Source
Edges used	Product-Category and Product-Source
Sample visualization	A sample of 100 products was drawn to keep the network readable

7. Network Analysis

The graph was analyzed using degree and betweenness centrality. Degree helps identify the most connected nodes, while betweenness centrality helps identify nodes that act as bridges between different parts of the graph.



Top node	Type	Degree / Centrality insight
eBay API	Source	Highest degree and highest betweenness because it collected the largest number of products.
Selenium	Source	Second strongest source node, also important because it provides rating and review values.
Shoes	Category	Most common category and one of the strongest category nodes.
Tops and Outerwear	Category	Highly connected categories with many products.
Bags, Modest Wear, Swimwear, Dresses	Category	Connected groups that support category-based exploration.

8. Visualizations and Heatmap

Several visualizations were created to make the collected data easier to understand. The charts show product distribution, ratings, reviews, prices, and category/source patterns. The heatmap was used to compare average price across categories and data sources.

Category Distribution Shows which women fashion categories appear most in the dataset.	Rating Analysis Calculates average rating from products where rating was available.	Review Analysis Uses review counts to identify highly reviewed products and categories.
Price Distribution Displays the spread of product prices and detects very high or low prices.	Heatmap Compares average price by category and source.	Network Graph Visualizes product links with categories and acquisition sources.

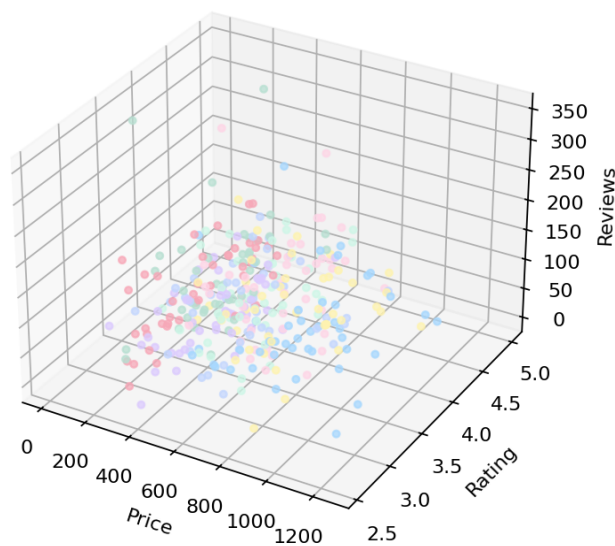
Important values from the notebook:

Metric	Value
Average price	388.78
Highest price	9000.0
Lowest price	0.99
Products with rating	1578
Average rating	4.0
Highest rating	5.0
Lowest rating	1.0
Products with reviews	1578
Total reviews	86534
Highest reviews	965

9. Point Cloud Analysis

For Phase 2, point cloud analysis is used to represent products in a 3D space. The main idea is to place each product according to three numerical features such as price, rating, and number of reviews. This makes it easier to notice clustering patterns, outliers, and groups of similar products.

3D Point Cloud: Price, Rating, and Reviews



Axis	Meaning
X-axis	Product price
Y-axis	Product rating
Z-axis	Number of reviews
Color/grouping	Product category
Purpose	Identify products with similar price-rating-review behavior and detect outliers

Since rating and review values were mainly available from Selenium/Amazon products, this part depends on products that have valid numerical rating and review values. Missing ratings should not be treated as wrong data; they are unavailable because some sources do not provide this field.

10. Final GUI

The final GUI was implemented in the notebook using ipywidgets and Plotly. It allows the user to filter the dataset by category and source, then immediately view the number of filtered products and important summary values. This supports fast exploration during the discussion.

GUI requirement	How it was handled
Scalable	Filters work on the full dataframe without changing the source data.
Usable	Dropdowns allow category and source selection without writing code.
Responsive	The summary output updates when filter values change.
Consistent	The same columns are displayed in the results table.
Accessible	Clear labels such as Category and Source make the interface easy to understand.
Reliable	The GUI uses the cleaned dataframe, so outputs match the final dataset.
Clear navigation	The user can choose All or a specific category/source.
Fast feedback	Summary values and top rows appear directly in the notebook output.
Intuitive interaction	The user interacts with dropdowns and interactive charts instead of editing code.

Extra interactive outputs:

Output	Purpose
Interactive category distribution	Shows product counts by category using Plotly.
Top 10 most reviewed products	Highlights products with the strongest review activity.
Filtered product table	Displays product name, category, source, price, rating, and reviews after filtering.

11. Conclusion

The project successfully implements the required e-commerce data acquisition and visualization pipeline for women fashion products. Data was collected using BeautifulSoup, the eBay API, and Selenium. The records were cleaned and combined into one dataset, then analyzed using charts, NetworkX graph analysis, a heatmap, point cloud analysis, and an interactive GUI.

The final dataset contains 5051 products from three sources and eight women fashion categories. Network analysis showed the most connected sources and categories, while visualizations helped explain price, rating, review, and category patterns. The GUI makes the final outcome easier to explore and supports the bonus requirement related to design quality and usability.

Submission Checklist

Item	Status
Complete source code notebook	Upload the full notebook/code file
Dataset files	Include girls_fashion_products.csv or any file needed to run the notebook
PDF report	This file
Data acquisition methods	BeautifulSoup, eBay API, and Selenium included
Phase 1 requirements	Collection, preprocessing, graph construction, visualization, heatmap
Phase 2 requirements	Point cloud analysis and final GUI
Submission timing	Upload 24 hours before the final discussion

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